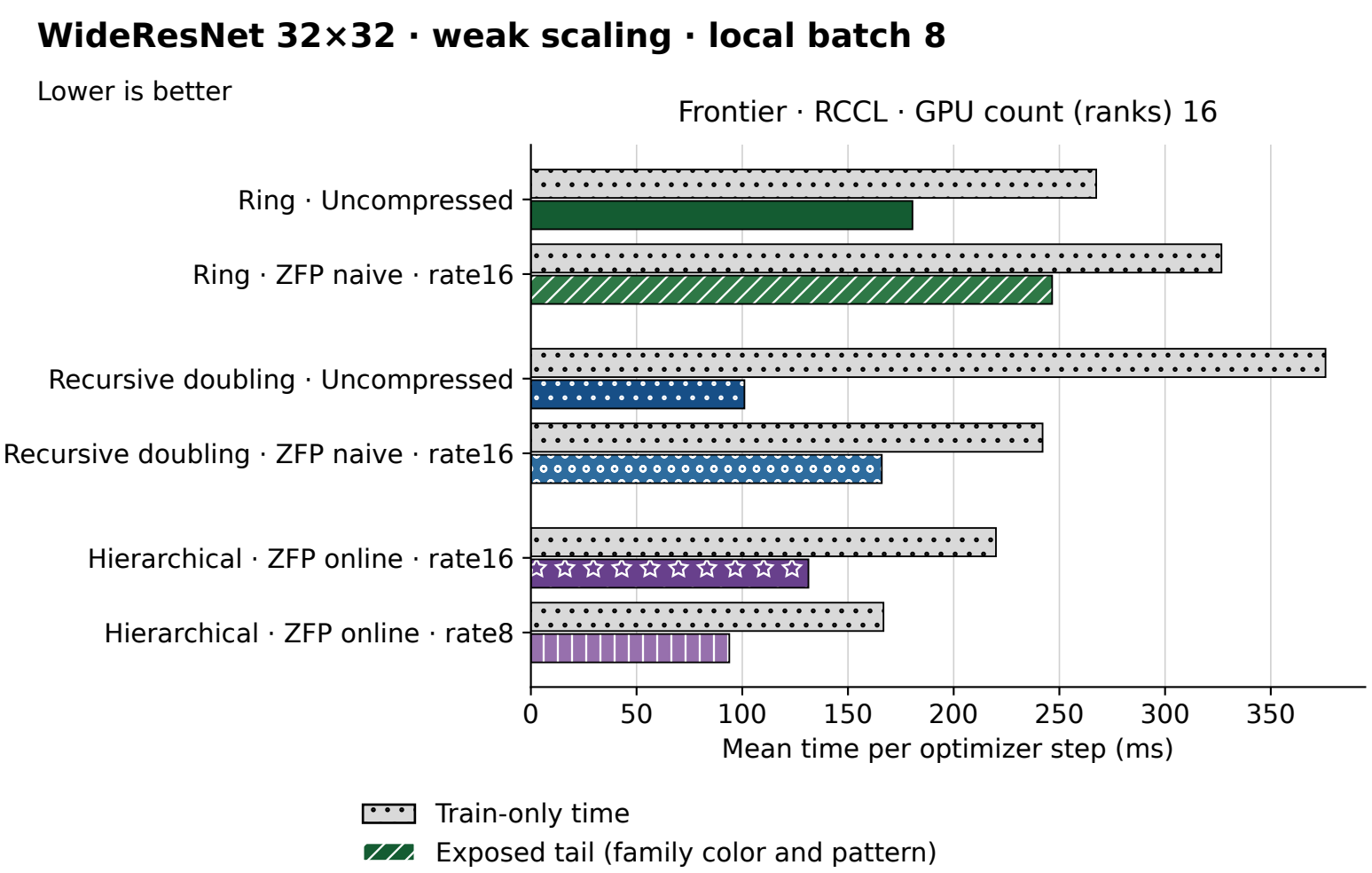

WideResNet 32×32 · weak scaling · local batch 8
Lower is better
Frontier · RCCL · GPU count (ranks) 16
Ring · Uncompressed
Ring · ZFP naive · rate16
Recursive doubling · Uncompressed
Recursive doubling · ZFP naive · rate16
Hierarchical · ZFP online · rate16
Hierarchical · ZFP online · rate8
0
50
100
150
200
250
300
350
Mean time per optimizer step (ms)
Train-only time
Exposed tail (family color and pattern)
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate